

Practice Test III (Version 2)

Course

- I) Write your name, the date, and the course title (i.e. Precalculus).
- II) Show all work.
- III) Draw a rectangle around your final answer for each problem.
- IV) A graphing calculator is required, but is not allowed for some problems.

① i) If "f" is an odd function, and $f(-18) = 4$, find $f(18)$.

ii) If "f" is an even function, and $f(0.3) = 10$, find $f(-0.3)$.

② The equation of a relation is written below. Write the equation of each reflection. Simplify if possible.

$$x = y^4$$

i) Reflect across the x-axis

ii) Reflect across the origin

Is the relation symmetric about the

iii) x-axis?

iv) origin?

③ A relation is written below in table form. Perform the following reflections and fill in the tables.

x	3	-2	-5	4	11
y	-7	14	-1	6	-2

i) Reflect across the y-axis

x					
y					

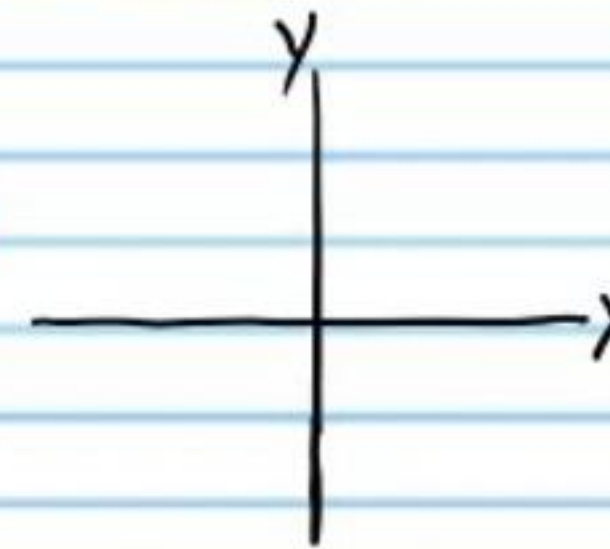
ii) Reflect across the origin

x					
y					

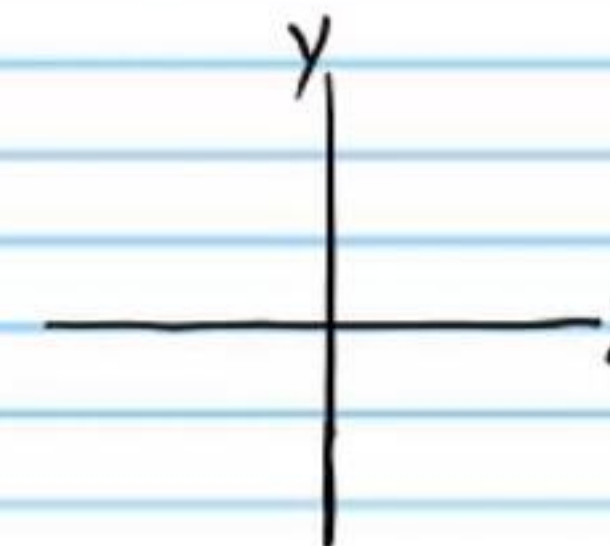
A relation is written below in the form of a graph. Draw the graphs of the indicated reflections.



iii) Reflect across the x-axis



iv) Reflect across the y-axis

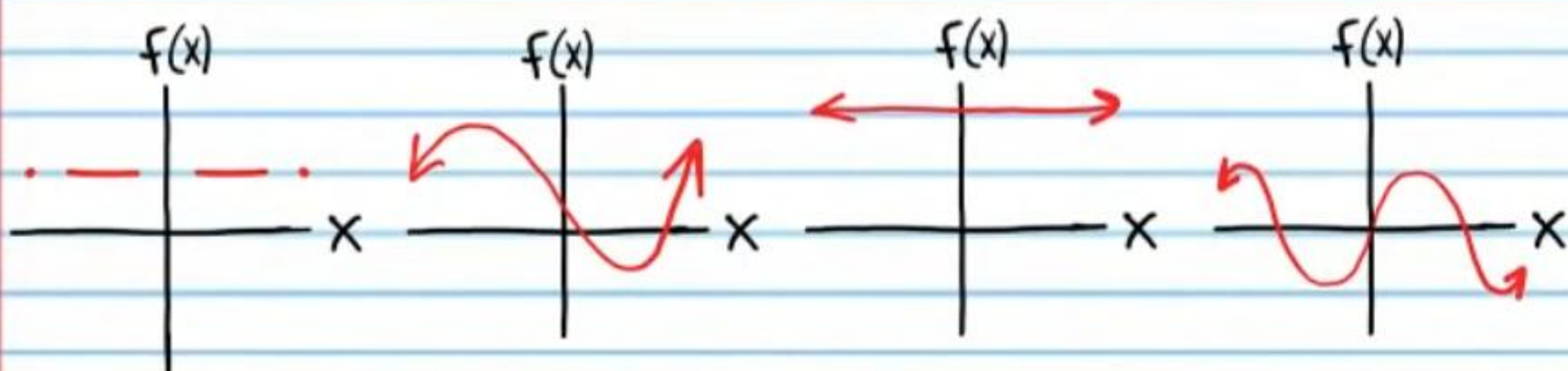
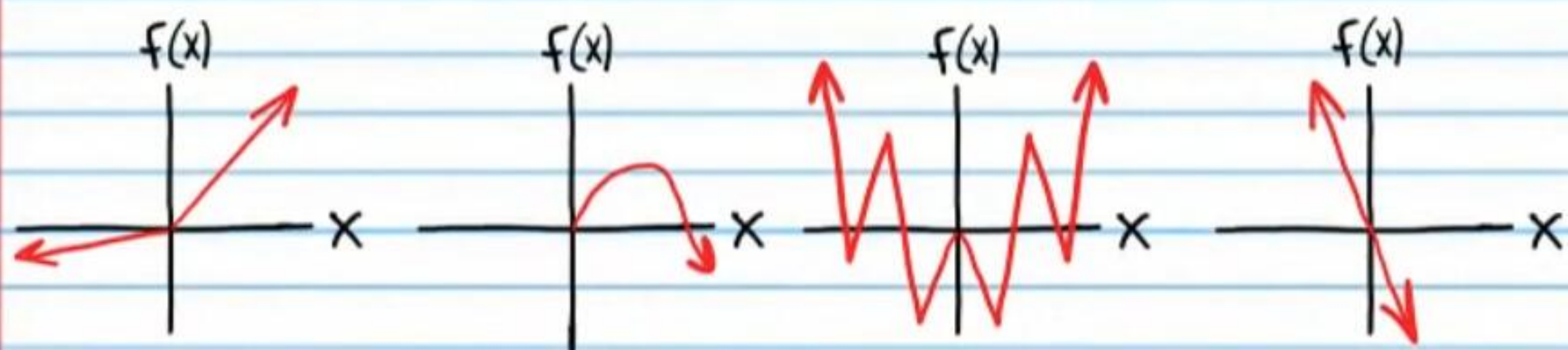


④ i) "f" and "g" are functions represented by the tables below. If "f" is an odd function and "g" is an even function, fill the empty spaces with other values of the functions.

x	-6	-8	-13	-10	4				
f(x)	-1	-2	36	26	8				

x	15	14	13	2	-7				
g(x)	-9	5	-3	2	-10				

ii) Determine if the graphs below are even functions, odd functions, or neither.



⑤ Each table represents a relation that has the indicated symmetry. Fill in the missing parts of the table.

i) Symmetric about the origin.

x	16	8	-3	-8	14										
y	3	-4	0	13	-2										

ii) Symmetric about the Y-axis.

x	17	16	15	-14	-16										
y	-2	-4	15	14	0										

⑥ Solve the following equations.

i) $\frac{3^6}{\sqrt[4]{3}} = \frac{27^x}{9^x}$

ii) $3 = 3 - \frac{\log_2 X}{8}$

⑦ i) Convert the equation below to rectangular form.
Note: Modifications will be necessary. Square both sides of the equation so that $\sqrt{x^2+y^2}$ is converted to x^2+y^2 in your final answer.

$$r = 1 - \tan \theta$$

ii) Determine if the function below is even, odd, or neither.

$$f(x) = \sin^2 x + \cos x$$

⑧ Relations are written below in various forms. Write each inverse relation in its corresponding form.

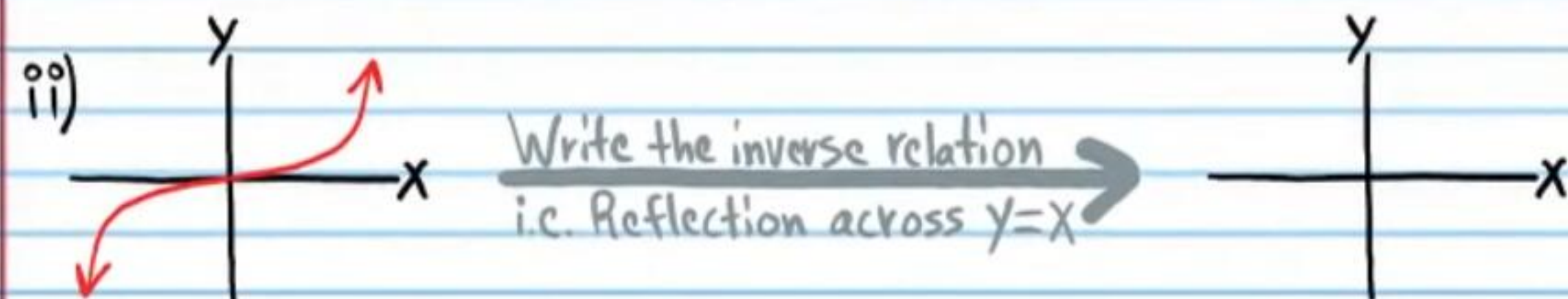
i)

x	87	32	-20	-41	17
y	33	-31	-4	-18	0

 Write the inverse relation $\rightarrow$

x					
y					

i.e. Reflection across $y=x$



iii) $\sqrt{x} = 1 - y$ Write the inverse relation $\rightarrow$
i.e. Reflection across $y=x$

iv) $\log_2 y^2 = x^2 - 1$ Write the inverse relation $\rightarrow$
i.e. Reflection across $y=x$

⑨ i) Create parametric equations for the relation below. Use "t" as the parameter.

$$x = y^2 + y - 1$$

ii) Determine if the function below is even, odd, or neither.

$$f(x) = x^2 \sin x$$

Convert the rectangular coordinates below to polar coordinates. Use degree units for angle measures. Round to the nearest thousandth if necessary.

* You may not use a calculator.

⑩ $(-2\frac{1}{2}, 0)$

⑪ $(-4, -4)$

⑫ $(2, -6)$

⑬ $(-4, 5)$

Convert the following polar coordinates to rectangular coordinates.
Round to the nearest hundredth if necessary.

* You may not use a calculator.

⑭ $(1, 240^\circ)$

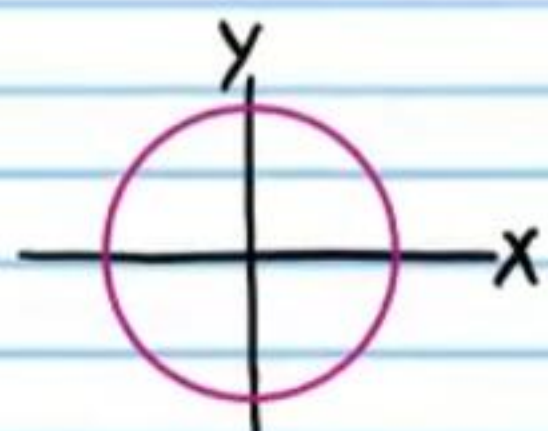
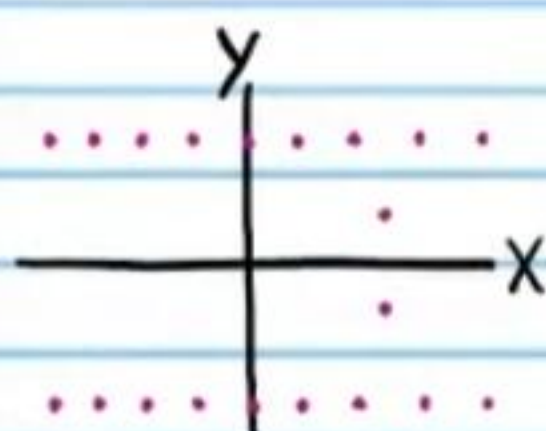
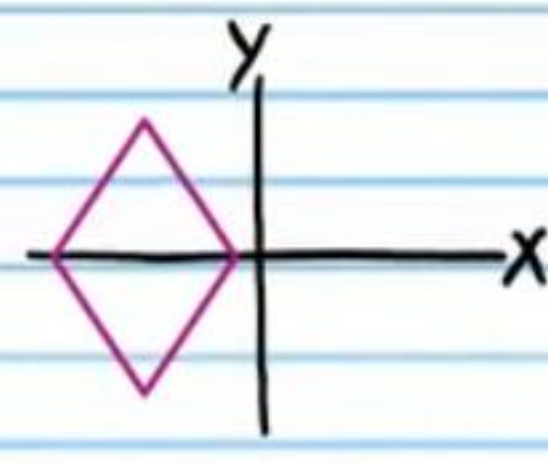
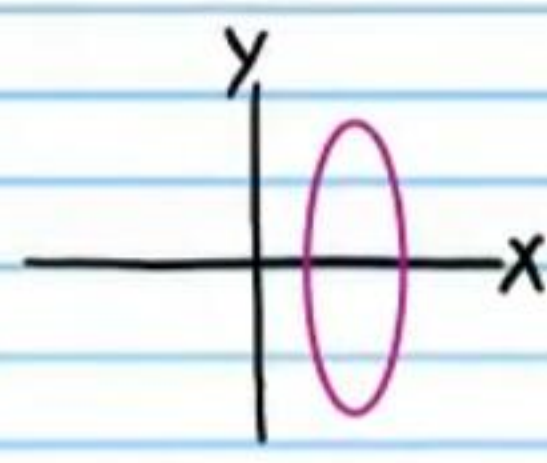
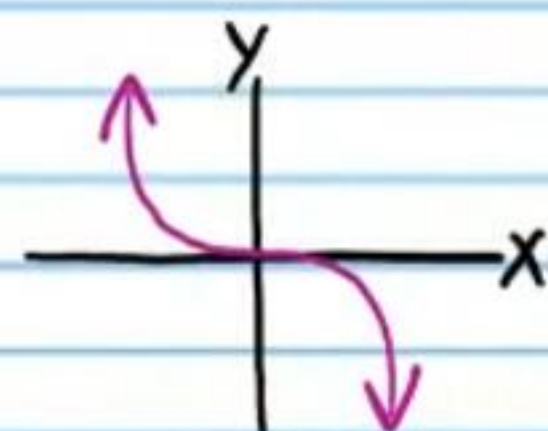
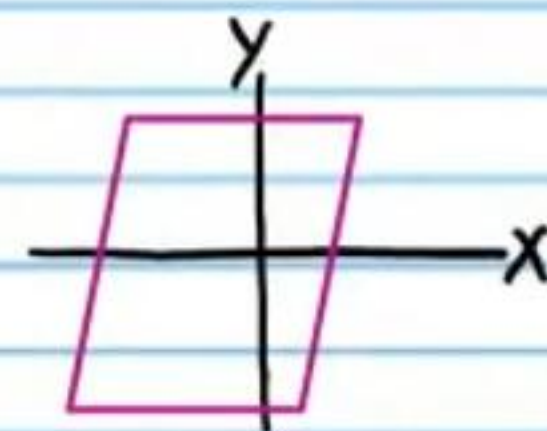
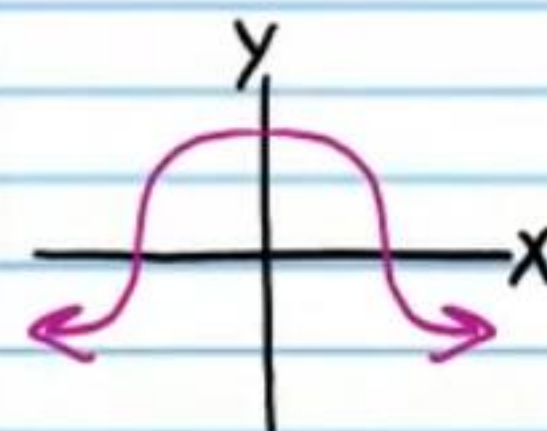
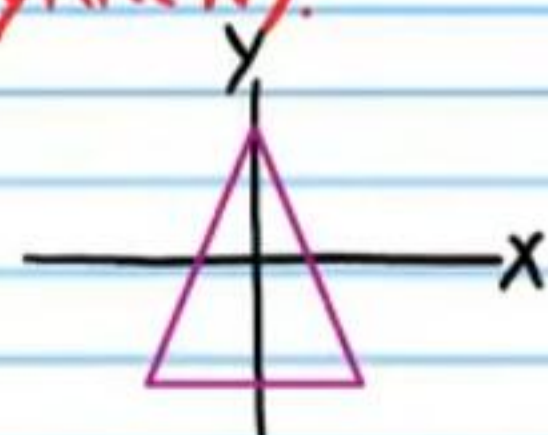
⑮ $(3, 180^\circ)$

⑩ $(3, 160^\circ)$

** You may not use a calculator.*

⑪ $(-4, -150^\circ)$

⑫ Relations are written below in the form of graphs. If a relation appears to be symmetric about the x-axis, write "x." If a relation appears to be symmetric about the y-axis, write "y." If a relation appears to be symmetric about the origin, write "O." If the relation shows no apparent symmetry, write "N." *Some graphs might display multiple types of symmetry.*

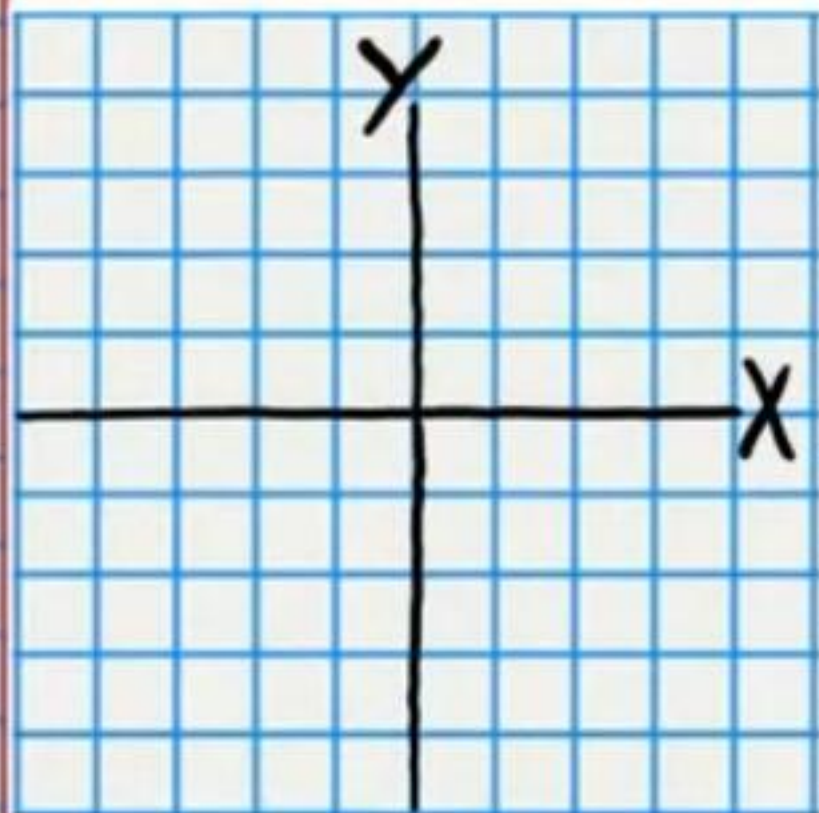


Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

⑱ $r = 1 - 3\sin\theta$

1st Point

2nd Point



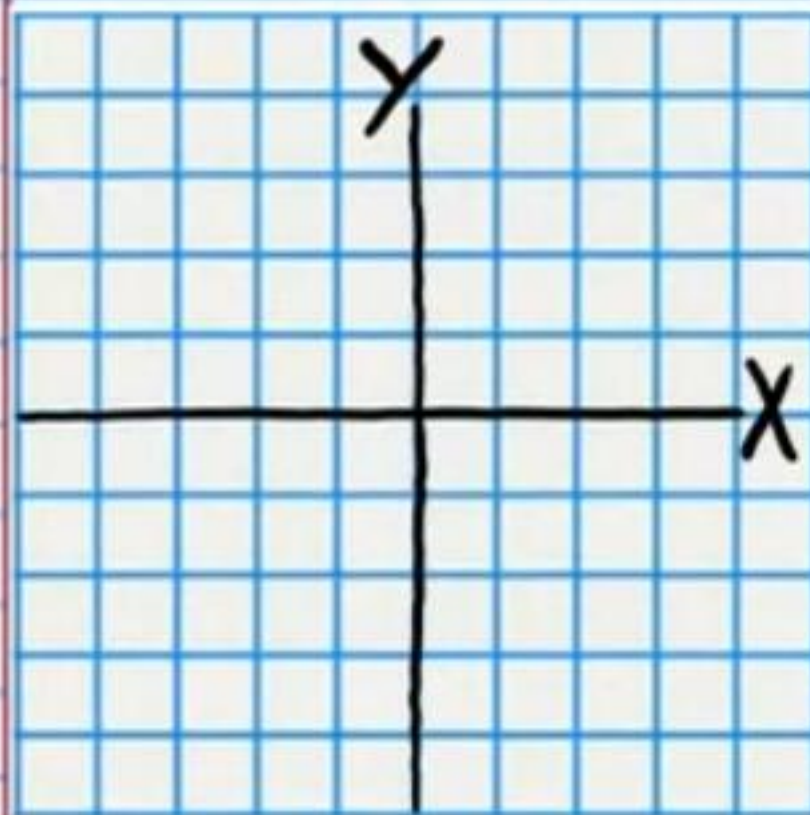
3rd Point

4th Point

⑳ $r = \frac{3}{1 + \sin\theta}$

1st Point

2nd Point

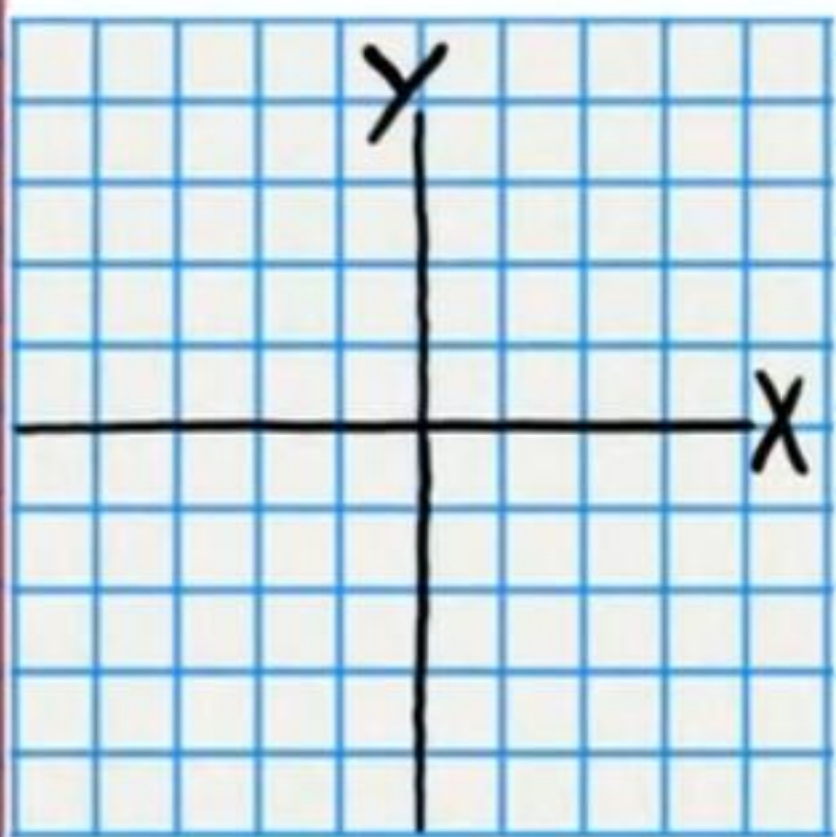


3rd Point

4th Point

②① $3 = r \cos \theta$

* It is not necessary to write the coordinates of points.



②② Convert the equation below to polar form. Simplify if possible.

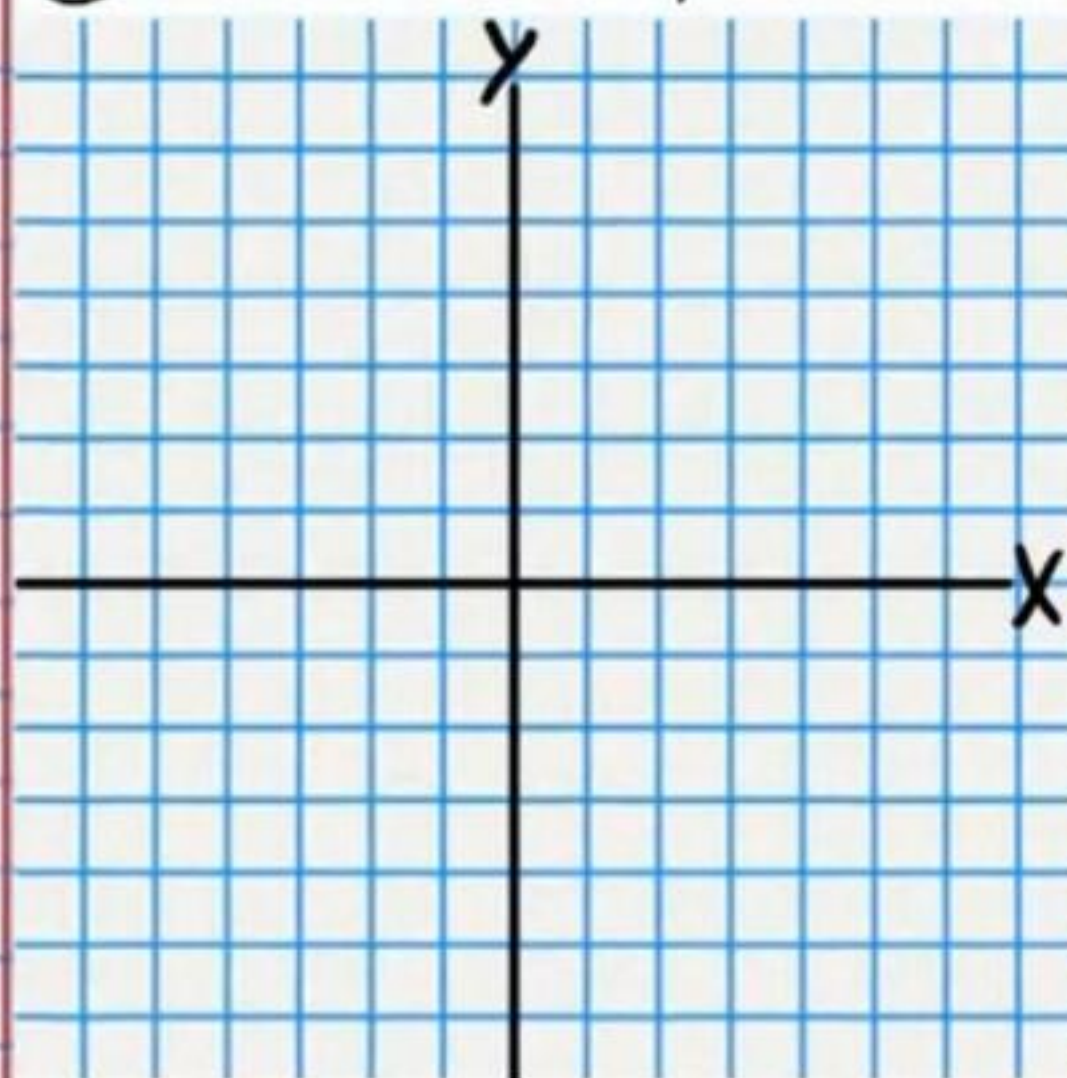
$$y = x$$

Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot. Then eliminate the parameter.

②③ $x = 5 \cos t$, $y = 5 \sin t$

Point #1

Point #2



Point #3

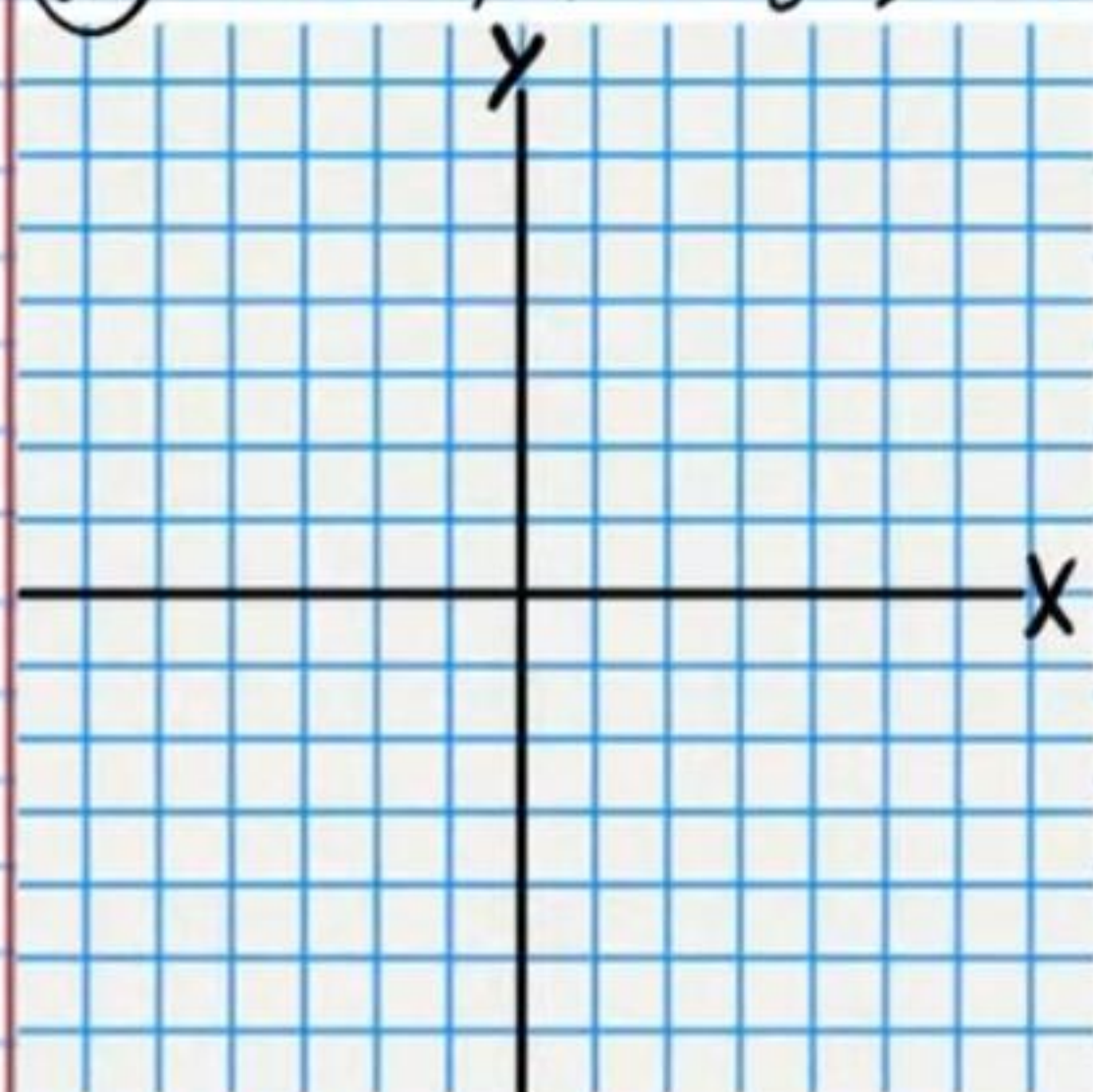
Eliminate the parameter

Point #4

②④ $x = t, y = \frac{1}{t-3}$

Point #1

Point #2



Point #3

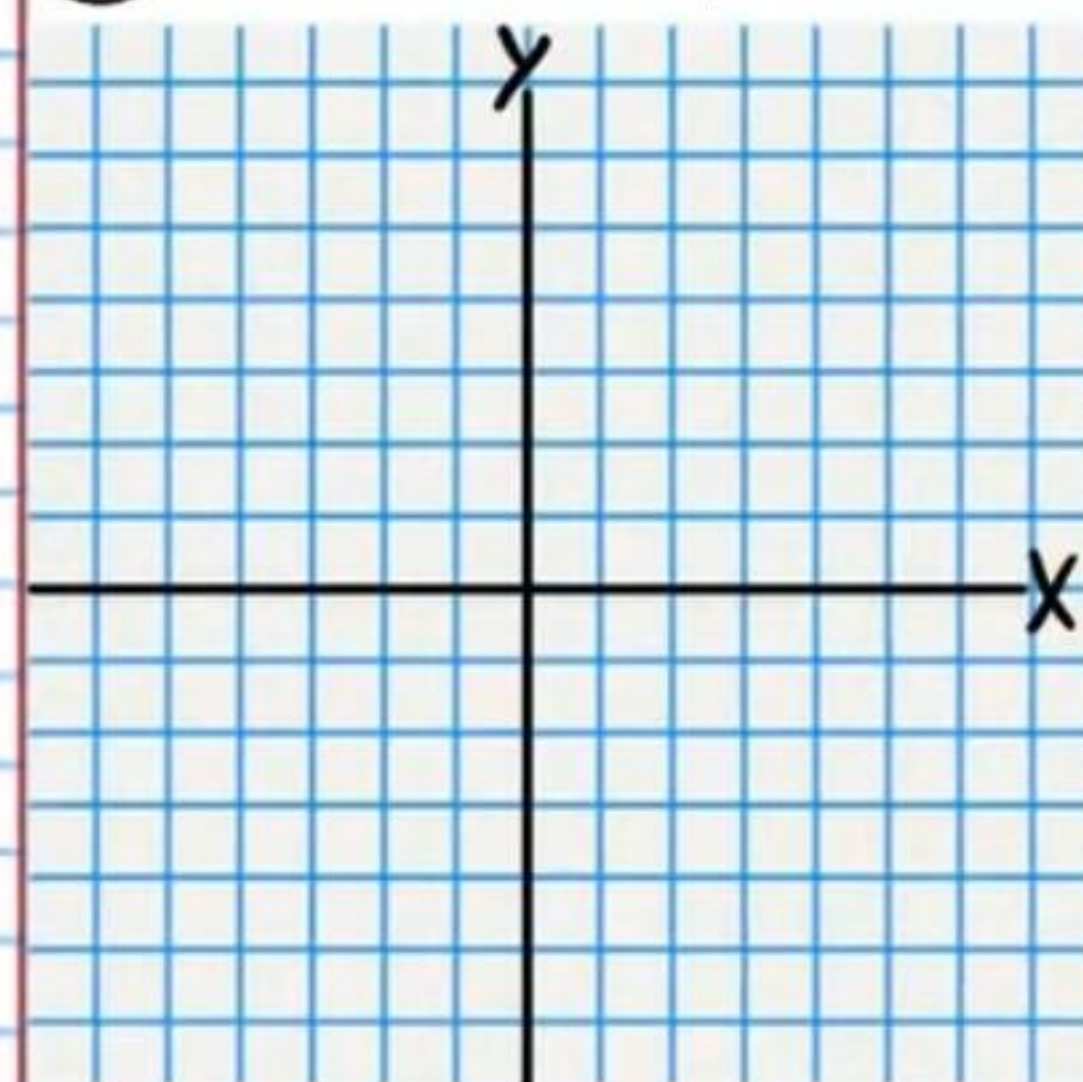
Eliminate the parameter

Point #4

②⑤ $x = -3^t + 2, y = 3^t$

Point #1

Point #2



Point #3

Eliminate the parameter

Point #4

Practice Test III (Version 2) Answers

① i) $f(18) = -4$

ii) $f(-0.3) = 10$

② i) $x = y^4$

ii) $-x = y^4$

iii) Yes

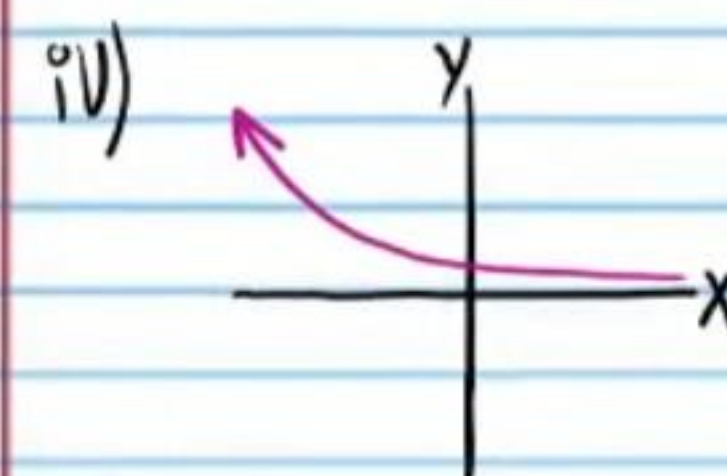
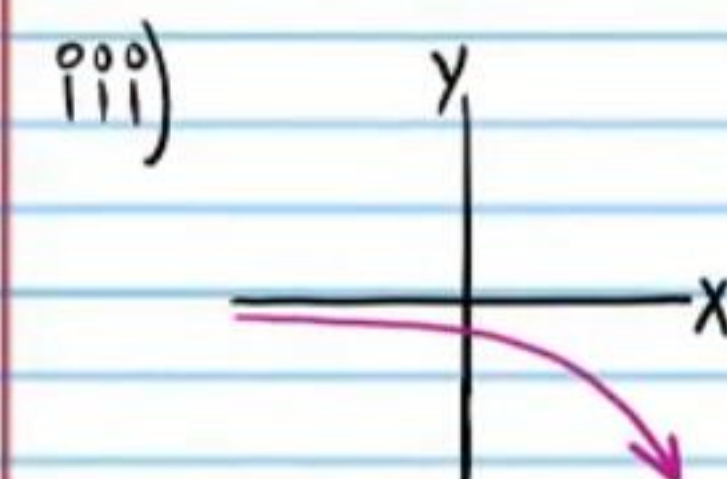
iv) No

③ i)

x	-3	2	5	-4	-11
y	-7	14	-1	6	-2

ii)

x	-3	2	5	-4	-11
y	7	-14	1	-6	2



④ i)

x	-6	-8	-13	-10	4	6	8	13	10	-4
f(x)	-1	-2	36	26	8	1	2	-26	-26	-8

ii)

x	15	14	13	2	-7	-15	-14	-13	-2	7
g(x)	-9	5	-3	2	-10	-9	5	-3	2	-10

iii) Neither Even Neither Even Odd
Even Neither Even Odd

⑤ i)

x	16	8	-3	-8	14	-16	-8	3	8	-14
y	3	-4	0	13	-2	-3	4	0	-13	2

ii)

x	17	16	15	-14	-16	-17	-16	-15	14	16
y	-2	-4	15	14	0	-2	-4	15	14	0

⑥ i) $x = 2^{3/4}$ ii) $x = 1$

⑦ i) $\sqrt{x^2 + y^2} = 1 - \frac{y}{x}$ This equation is lacking corresponding values and thus is not acceptable.

$x^2 + y^2 = \left(1 - \frac{y}{x}\right)^2$

ii) $f(-x) = \sin^2(-x) + \cos(-x)$

$f(-x) = [\sin(-x)]^2 + \cos x$

$f(-x) = [-\sin x]^2 + \cos x$

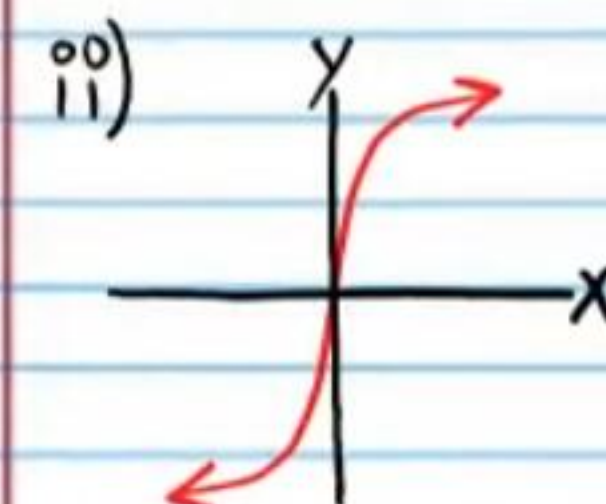
$f(-x) = \sin^2 x + \cos x$

$f(-x) = f(x)$

Even

⑧ i)

x	33	-31	-4	-18	0
y	87	32	-20	-41	17



iii) $\sqrt{y} = 1 - x$

iv) $\log_2 x^2 = y^2 - 1$

⑨ i) $y = t, x = t^2 + t - 1$

ii) $f(-x) = (-x)^2 \sin(-x)$

$f(-x) = x^2 (-\sin x)$

$f(-x) = -x^2 \sin x$

$f(-x) = -f(x)$

Odd

⑩ Other answers are possible.

$(-2\frac{1}{2}, 0^\circ)$ or $(2\frac{1}{2}, 180^\circ)$

or $(2\frac{1}{2}, -180^\circ)$

⑪ Other answers are possible.

$(4, 225^\circ)$ or $(4, -135^\circ)$

or $(-4, 45^\circ)$

⑫ Other answers are possible.

$(2\sqrt{10}, -71.565^\circ)$ or $(2\sqrt{10}, 288.435^\circ)$

or $(-2\sqrt{10}, 108.435^\circ)$

⑬ Other answers are possible.

$(\sqrt{41}, 128.66^\circ)$ or $(-\sqrt{41}, -51.34^\circ)$

⑭ $(-\frac{1}{2}, -\frac{\sqrt{3}}{2})$

⑮ $(-3, 0)$

⑯ $(-2.82, 1.03)$

⑰ $(2\sqrt{3}, 2)$

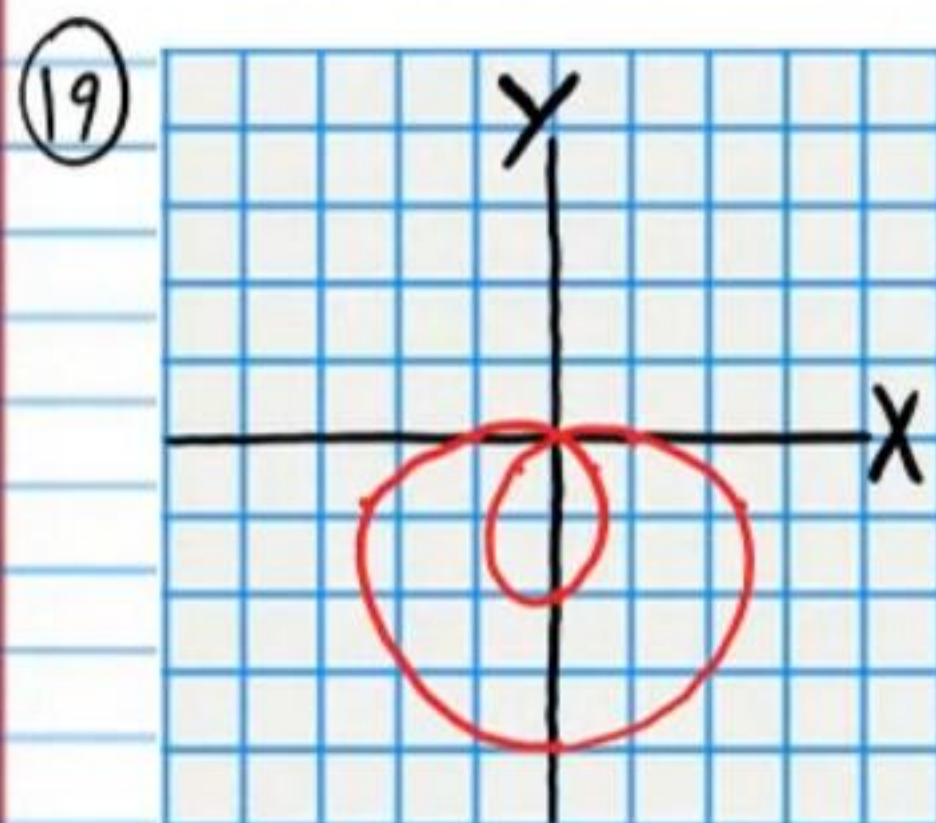
⑱

x	N	y
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y	N	0
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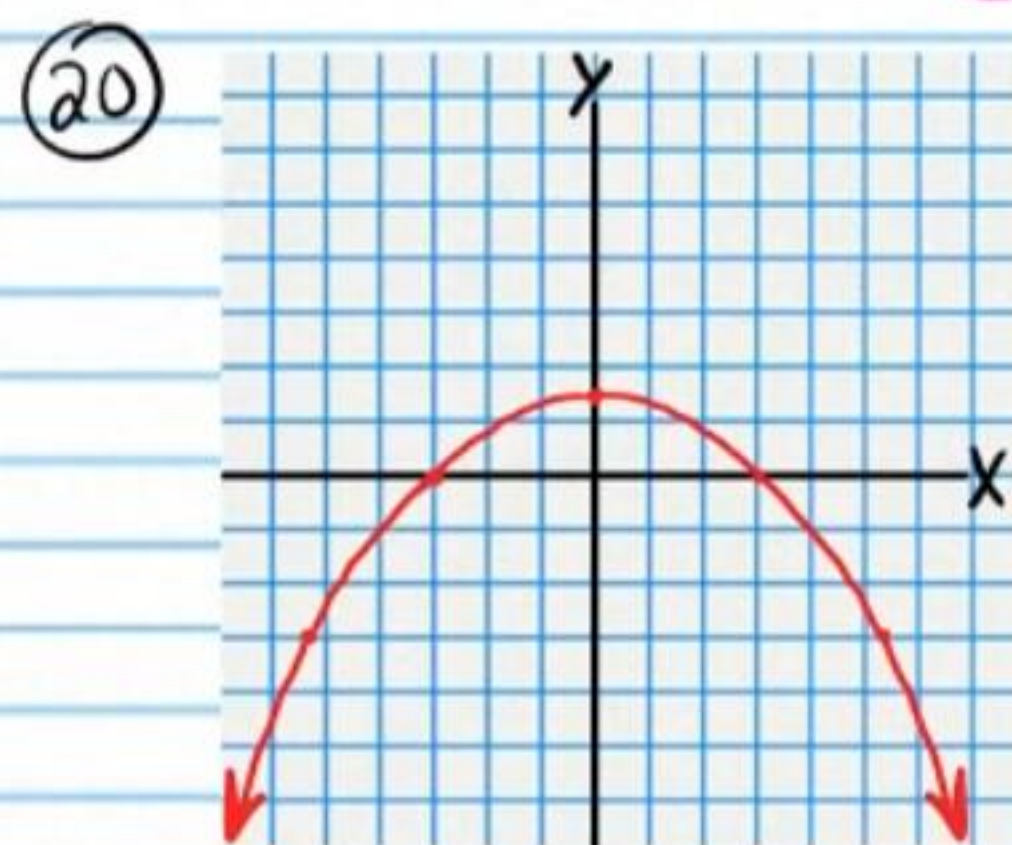
N	x	x
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x	0	x & y & 0
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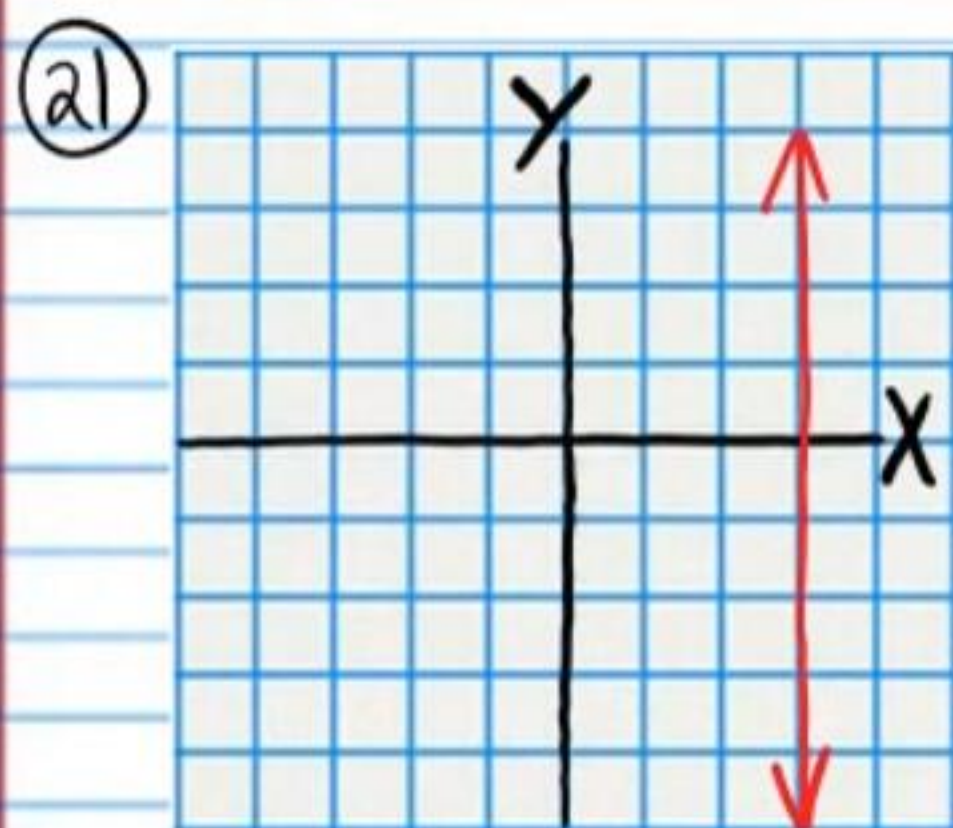
Other points are possible.
Only 4 are necessary.

$$\begin{array}{l} (1, 0^\circ) \quad (-2, 90^\circ) \quad (4, -90^\circ) \\ (1, 180^\circ) \quad (-\frac{1}{2}, 30^\circ) \quad (2\frac{1}{2}, -30^\circ) \\ (4, 270^\circ) \end{array}$$

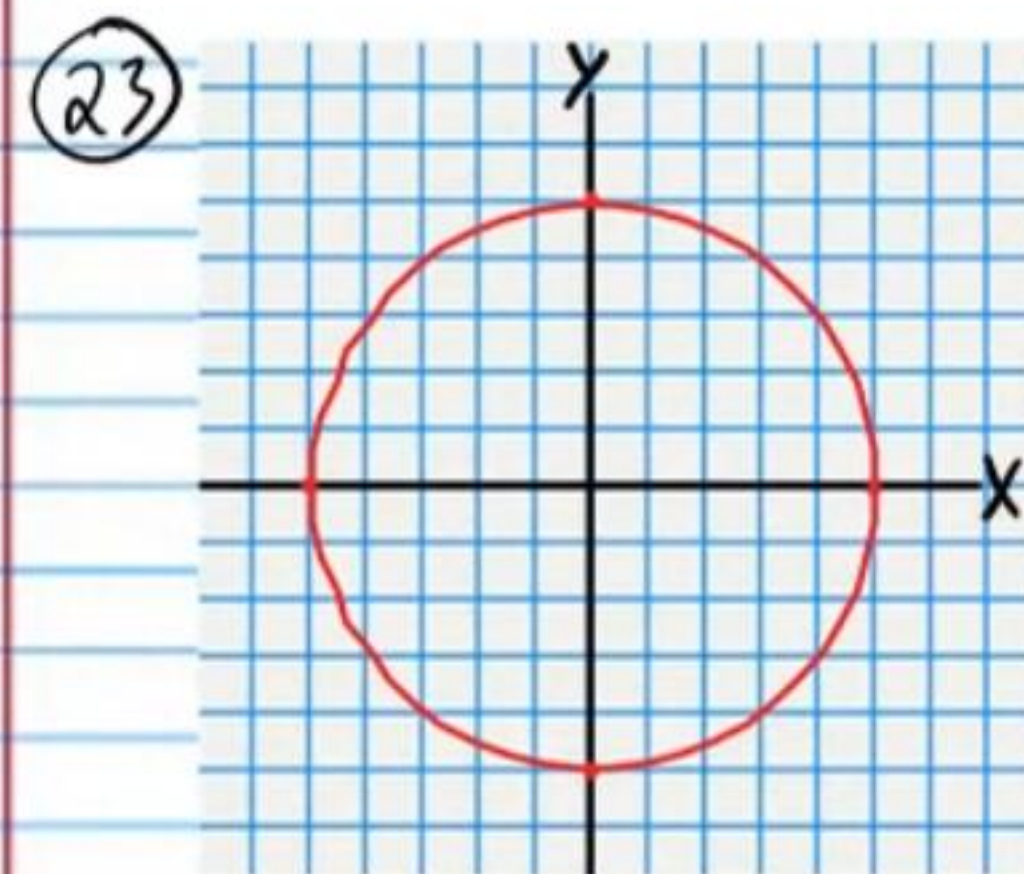


Other points are possible.
Only 4 are necessary.

$$\begin{array}{l} (3, 0^\circ) \quad (3, 180^\circ) \quad (\frac{3}{2}, 90^\circ) \\ (3, -180^\circ) \quad (2, 30^\circ) \quad (6, -30^\circ) \end{array}$$



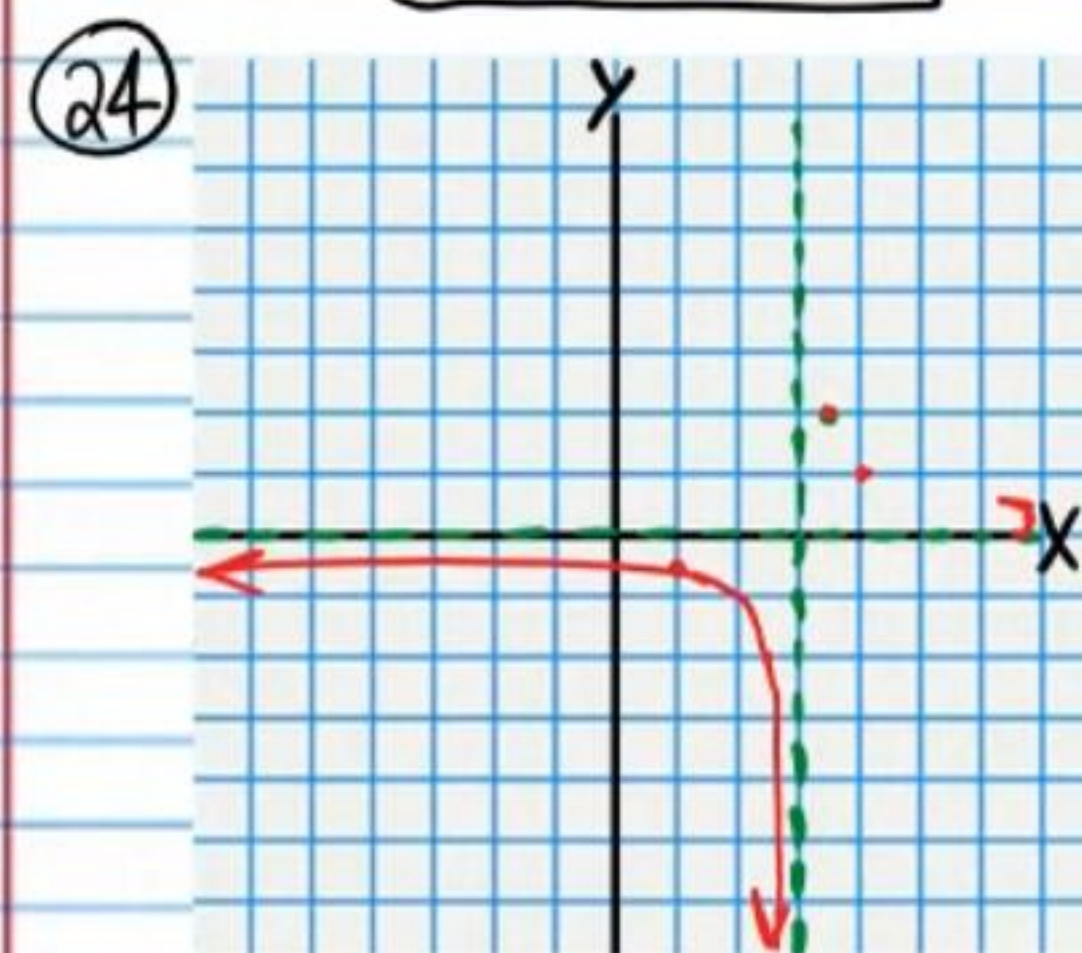
22) $r \sin \theta = r \cos \theta$ or
 $\sin \theta = \cos \theta$ or
 $\tan \theta = 1$ or
 $\cot \theta = 1$ or
 $\theta = 45^\circ$ or
 $\theta = 225^\circ$ or
 $\theta = -135^\circ$



Other points are possible.
Only 4 are necessary.

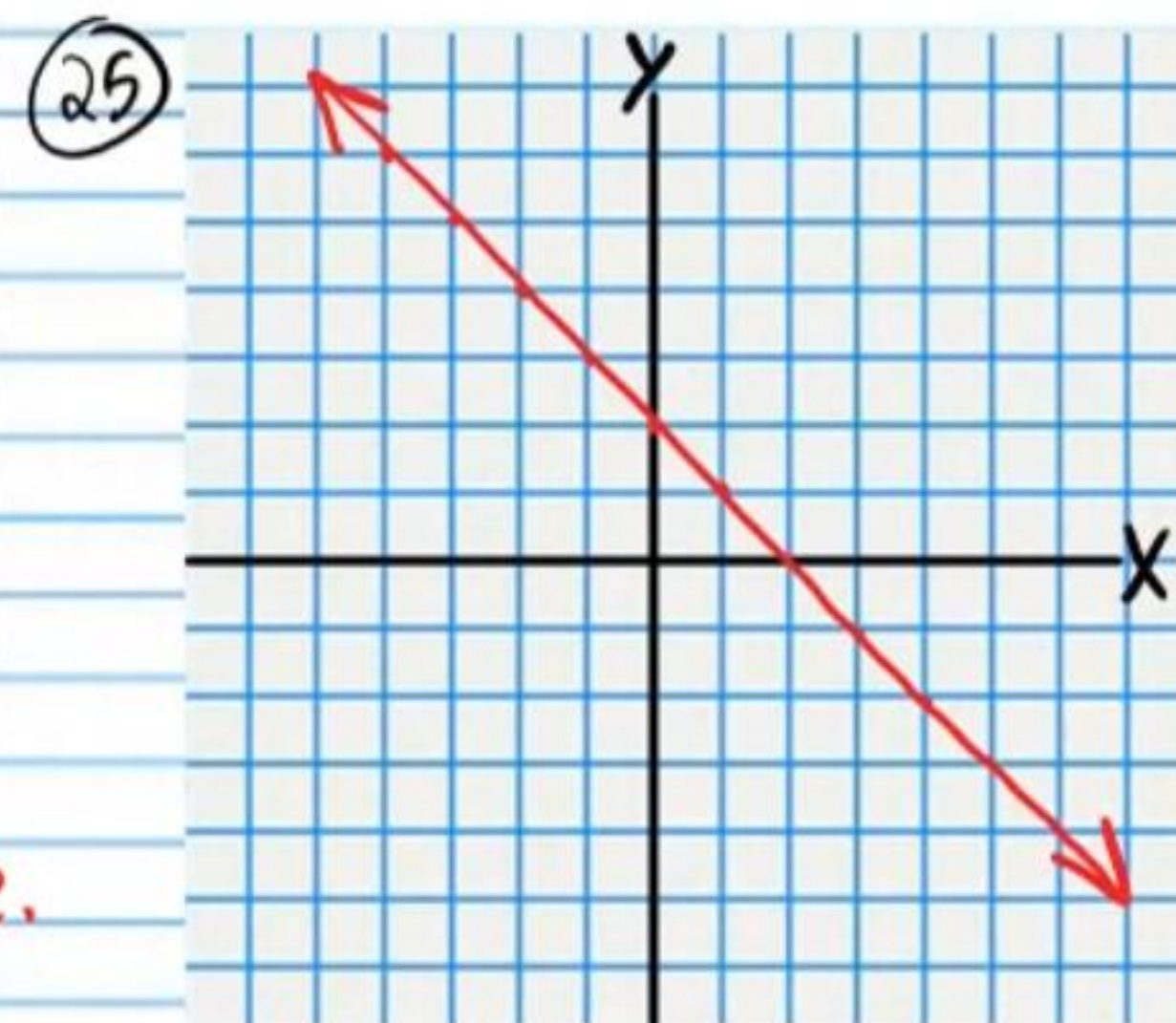
$$(0, 5) \quad (5, 0) \quad (0, -5) \quad (-5, 0)$$

$$x^2 + y^2 = 25$$



Other points are possible.
Only 4 are necessary.

$$\begin{array}{l} (2, -1) \quad (4, 1) \quad (1, -\frac{1}{2}) \quad (5, \frac{1}{2}) \\ y = \frac{1}{x-3} \end{array}$$



Other points are possible.
Only 4 are necessary.

$$\begin{array}{l} (1, 1) \quad (-1, 3) \quad (1\frac{2}{3}, \frac{1}{3}) \\ (1\frac{8}{9}, \frac{1}{9}) \quad x = -y + 2 \end{array}$$